

END TERM EXAMINATION**THIRD SEMESTER [B.TECH] DECEMBER 2025 – JANUARY 2026****Paper Code: PC-205****Subject: Object Oriented Programming Using Java****Time: 3 Hours****Maximum Marks: 60****Note:** Attempt all questions as directed. Internal choice is indicated within units.

Q. No.	Question	Marks
1	Answer all the following questions: (a) Explain different types of nested classes in Java. (b) Explain multitasking and multithreading in Java. (c) Discuss wrapper classes and adapter classes. (d) Explain the life cycle of an Applet. (e) What is the Java Native Interface (JNI)?	5 × 4 = 20
UNIT – I		
2	Explain fundamental features of Java and how it achieves architecture neutrality and robustness.	10
3	[OR] Explain the different runtime memory data areas in the Java Virtual Machine (JVM) architecture.	10
UNIT – II		
4	Explain the keywords try, catch, throw, throws, and finally. Write a Java program which catches two different types of exceptions when the program is executed twice with varying inputs.	10
5	[OR] Explain the concept of Dynamic Method Dispatch. Write a Java program illustrating dynamic method dispatch through the use of an abstract class and derived classes to compute the volume of a cuboid and a cube.	10

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Q. No.	Question	Marks
UNIT – III		
6	What is a Layout Manager? Explain any five standard Layout Managers in Java AWT / Swing with their respective constructors and behavior.	10
7	[OR] Write a program using Java Swing which displays four push buttons and a label. Each button displays an icon that represents the flag of a country. When a button is pressed, the name of that country is displayed in the label.	10
UNIT – IV		
8	Explain the different types of JDBC drivers (Type-1 to Type-4). Write a complete Java program to fetch and display all records of a database table using Java Database Connectivity (JDBC).	10
9	[OR] Explain how Remote Method Invocation (RMI) works in Java, including the roles of Stub, Skeleton, and RMI Registry. Write code to illustrate remote method invocation.	10

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